



PROFESSIONAL
EDUCATION



Digital Plus⁺
Programs

```
elif _operation == "MIRROR_Z":  
    mirror_mod.use_x = False  
    mirror_mod.use_y = False  
    mirror_mod.use_z = True
```

```
mirror_ob.select= 1  
modifier_ob.select=1  
bpy.context.scene.objects.active = modifier_ob  
print("Selected" + str(modifier_ob)) #  
#mirror_ob.select = 1
```

Data

Leadership:

Transforming the Corporation's
Operations, Management, and
Mindset to Leverage Data, AI,
and Cloud Computing



**Business leaders don't need
to be IT engineers or data
scientists; what is necessary
is for them to understand the
fundamentals and how new
tools can be used to add
value to a business**

Dr. Abel Sanchez

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Breaking Down the Science of Data



This MIT Professional Education course will increase your ability to extract valuable information from raw data and make informed decisions based on it. Aimed at providing knowledge of the data ecosystem, this course relates to the most important tools such as databases, artificial intelligence, data governance, DevOps, cloud, blockchain, machine learning, and cybersecurity.

The data leadership role encompasses a reconstruction of the organization's operations.

This course gives you an understanding of the technologies that are driving companies into an uncertain yet exciting future filled with new opportunities.

Discover how data leadership can **reconstruct your organization's culture, transform the organization of information and data, and guide you towards implementing new technologies to better manage your business.** Determine the best way for your employees to integrate groundbreaking tools **to facilitate their daily responsibilities.**

Enroll now in MIT Professional Education's Data Leadership course and explore the advancements in technology, particularly in data storage.

The Worldwide Importance of Data Leadership



60%

of corporate data is stored in the cloud.

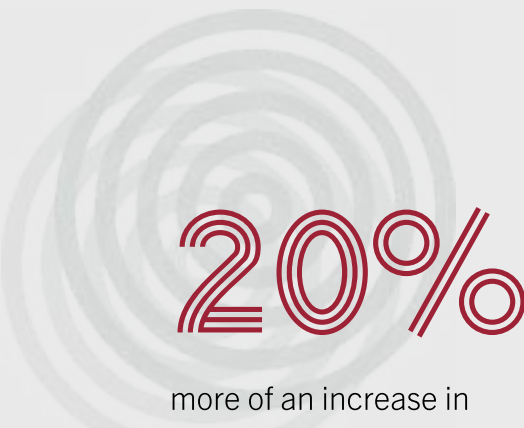
Source: Statista



38%

of CEOs say they champion their companies' analytics agendas, but only 9% of all other C-suite executives agree.

Source: McKinsey & Company



20%

more of an increase in revenue before taxes was achieved by dataanalytic firms than by non-data analytic firms.

Source: McKinsey & Company

An Online Program Offering a Profound Approach to Data Leadership



Surviving in a volatile, uncertain, complex, and ambiguous (VUCA) world calls for drastic change and adaptability. There's so much data, and often, no one knows what to do with it. The businesses who know what to do with massive amounts of data are—and will continue to be—top leading companies. They're moving thousands of times faster than others.

Effective business decisions depend on precise forecasting. It entails the ability to preemptively enact changes in a market constantly in flux and respond to increasingly shifting customer demands. Data-driven executive decision-making relies on the insights of advanced analytical techniques.

The Data Leadership course from MIT Professional Education aims to ideate new ways to use data, developing a holistic enterprise data strategy. This course analyzes how to make relevant data accessible to decision-makers at every level of the organization.



8 weeks



8-10 hours a week



Online

Key Takeaways

By the end of the course, participants will:

- 01** Comprehend the history and role of **data and the cloud** from its beginnings to now
- 02** Understand what **data platforms** are, how they work, and their applications
- 03** Leverage organizations' data infrastructures
- 04** Have the option to **build a simple database using the SQLite database engine**
- 05** Leverage existing company data for **success** and derive value from dormant data
- 06** Envision and practice precise decision-making based on data
- 07** Apply best practices in data governance and cybersecurity
- 08** Understand how the **principles of Lean DevOps can be applied** to optimizing data systems for organizational advancement
- 09** Understand how **cloud, machine learning, and data store technologies improve modern data systems'** speed and scaling processes
- 10** Discover a range of **frameworks designed to enhance collaboration** within your organization, ultimately boosting efficiency levels.
- 11** Gain a solid understanding of the essential principles for building data organization.



Participant Profile

Management and organizations who need to integrate and connect their data sources and databases to better leverage their existing data will benefit from this course too. Therefore, this course is primarily for:

Management level professionals and executives looking to champion a data-driven approach in their organizations to be able to make informed decisions.

More established organizations as well as younger start-ups seeking to improve data management and processing.

Professionals and executives in data management roles Chief of Data Analytics, Chief of Data, Chief of Machine Learning.

Organizations with a medium to high number of databases and medium-high data volumes that need to integrate and connect their sources with their databases to better leverage their use.

Other roles with significant responsibility wanting to improve their data analytics and optimize their utilization.

Organizations that operate with legacy systems, such as SAP, need to update their data structure.



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Our Professional Alumni Experience



Dharmendra Jain

Director of Operations at Kantar

This course has given me a holistic understanding and knowledge of data driven initiatives/projects. I learnt about the latest technologies such as data pipelines, low-code/no-code, CI/CD, cloud deployment, data compliance, and how I can deploy them. I find this very good for someone who already has some technical knowledge and is currently at the senior level and has the responsibility of leading data driven initiatives.

David Sánchez

Senior Advanced Support Engineer at Oracle

The toolbox provided in the course informs your decision when it comes to designing your business for digital. Besides, it warns you to keep your eyes open to evolution in technology to leverage it as you create and capture value. I cannot imagine a business thrive without putting into practice the lessons I've learnt.

Guillermo Valdivia

Manager, CIO Toolbox Engineering & Data Insights at IBM

This is such an important topic and to be able to take it right at the same time as ChatGPT came out is a differentiator in my career. I am able to have discussions and lead my team with proper guidance with the art of the possibilities in data science.

Online course **Data Leadership**: Transforming the Corporation's Operations, Management, and Mindset to Leverage Data, AI, and Cloud Computing

Instructors



Dr. Abel Sanchez

Executive Director of MIT's Geospatial Data Center, Research Scientist; Center for Complex Engineering Systems, Sociotechnical Systems Research Center, under the Schwarzman School of Computer Science

Dr. Sanchez holds a Ph.D. from the Massachusetts Institute of Technology (MIT). He is the Executive Director of MIT's Geospatial Data Center (GDC). His areas of specialty include the Internet of Things (IOT), Big Data, Cybersecurity, and Digital Innovation. He teaches graduate courses in Data Science, Cybersecurity, and Innovation. For the past eight years his research has focused on architecting large-scale computation.

In IOT, Dr. Sanchez led the global network architecture for the Internet of Things at MIT. The design addresses large-scale computation. Compared to the largest numbers in the world, 2 billion computers, 7 billion phones, and 7 billion people, IOT is orders of magnitude bigger. In similar work,

Dr. Sanchez directed simulation of the U.S. critical infrastructure with the National Infrastructure Simulation and Analysis Center (NISAC).ia.

In data science, early work included supply chain information engineering, analytics, simulation, and visualization with Wal-Mart, Kraft, and SAP. Dr. Sanchez extended his work to global anti-counterfeiting efforts with Johnson & Johnson, SAP, and Altria. Helping Altria scale track-and-trace using the Electronic Product Code Information Services standard from RFID. The work produced a global infrastructure used by industry and government to this day.

In enterprise computing, Dr. Sanchez led the design of a global data infrastructure simulator, modeling follow-the-sun engineering, to evaluate the impact of competing architectures on the performance, availability and reliability of the system for Ford Motor Company. The simulator modeled user actions, applications, background processes, network load, servers, storage, and global data centers. The work identified data center reductions opportunities estimated at a billion dollars in savings.

In cyber security, Dr. Sanchez directed impact analysis of large-scale cyber-attacks designing Cyber Ranges for the Department of Defense (DOD). Conducting repeatable experiments in impact analysis and the ability to model the cyber environment in a highly portable fashion. Looking at the insider threat Dr. Sanchez led the DOD security study on Enterprise Resource Planning systems across the United States Armed Forces.

In password security, Dr. Sanchez led the design of a password firewall (negative authentication) for the Intelligence Advanced Research Projects Activity (IARPA) agency. The Negative Filtering or Negative Authentication (NA) approach utilizes a form of complement profiles which resembles the censoring and maturation process of T- cells in the immune system.

In machine learning, addressing financial fraud, Dr. Sanchez designed a situational awareness framework that exploits different perspectives of the same financial data and assigns risk scores to entities (e.g. payment documents) to improve false positive ratios and assist the identification of fraudulent activity in huge and unlabeled financial data in collaboration with Accenture.

In physical security, Dr. Sanchez is developing algorithms to assess risk in the integration of information technology (IT) and operations technology (OT).

Dr. Sanchez is the founder and Chief Software Architect of the Open Source RFID platform project. Dr. Sanchez' software systems are used by Samsung, NEC, NTT, Hitachi, Motorola, SAP, IBM, and Microsoft. Other software initiatives are in use by Sandia National Laboratories, MIT, and by several organizations in East Asia and Europe.

Dr. Sanchez advises companies in the Americas, Europe, Middle East, and Asia.

“We cannot bet on the artificial intelligence, or the human being separate. We must get the fusion of both.”

Affiliations

- MIT Geospatial Data Center (GDC)
- MIT Sociotechnical Systems Research Center (SSRC)
- MIT Institute for Data, Systems, and Society (IDSS)
- MIT Center for Complex Engineering Systems (CCES)
- MIT Center for Computational Science and Engineering (CCSE)
- MIT Consortium for Improving Critical Infrastructure Cybersecurity (IC3)
- MIT Schwarzman School of Computer Science
- MIT AutoID Laboratory

Areas of Interest and Expertise

- Machine Learning
- Cyber/Physical Security
- Enterprise Computing
- Data Science
- IOT

Courses Taught with MIT Professional Education

- Applied Generative AI for Digital Transformation
- Blockchain: Disruptive Technology
- Cloud & DevOps: Continuous Transformation
- Data Leadership: Transforming the Corporation's Operations, Management, and Mindset to Leverage Data, AI, and Cloud Computing
- Digital Transformation: From AI and IOT to Cloud, Blockchain, and Cybersecurity



Prof. John R. Williams

Professor of Information Engineering, Civil and Environmental Engineering and Director of MIT Geospatial Data Center, and a faculty member in the Center for Computational Science and Engineering part of the Schwarzman School of Computer Science

John Williams holds a BA in Physics from Oxford University, a MS in Physics from UCLA, and a Ph.D. in Numerical Methods from University of Wales, Swansea. His research focuses on the application of Exascale computation to problems in cyber-physical systems, security and energy. His work on fault-tolerant computing using container migration won the IEEE, High Performance Extreme Computing Conference award for Best Innovation in 2019. He is director of MIT's Geospatial Data Center and from 2006-2012, was Director of the MIT Auto-ID Laboratory, where the Internet of Things was invented.

He is author or co-author of over 250 journal and conference papers, as well as the books on Rock Mechanics and RFID Technology.

He contributed to the 2013 report for the UK Office for Science Foresight Project- The Future of Manufacturing. Alongside Bill Gates and Larry Ellison, he was named as one of the 50 most powerful people in Computer Networks. He consults to companies including Accenture, Schlumberger, Shell, Total, Exxon, SAP Research, Microsoft Research, Kajima Corp, US Lincoln Laboratory, Sandia National Laboratories, US Intelligence Advanced Research Projects Activity, Motorola, Phillip-Morris Inc., Ford Motor Company, Exxon-Mobil, Shell, Total, and ARAMCO.

His international collaborations include Oxford and Cambridge Universities, HKUST, KACST, Alfaisal University, PolyU Hong Kong, Imperial College of Science and Technology UK, Malaysia University of Science and Technology (MUST), and Masdar Institute of Science and Technology Abu Dhabi. He organized the first Cyber-Physical Security Conference in the UK (2011) and along with Dr. Sanchez, he runs the MIT Applied Cyber Security Professional Education summer course. At MIT he teaches courses Architecting Software Systems (MIT 1.125) and Engineering Computation and Data Science (MIT 1.00/1.001).

In data engineering and data science, early work included simulation of Ford's global network, analysis of SAP smart grid billing system. For Altria, he analyzed the performance of item level tagging and also their implementation of an anti-counterfeiting system using the Electronic Product Code (EPC).

In password security, Dr. Williams was PI that developed the algorithms for a negative password authentication system for the Intelligence Advanced Research Projects Activity (IARPA) agency.

Dr. Williams advises companies in the Americas, Europe, Middle East, and Asia.

“Once you climb that first mountain of knowledge, you will be able to teach yourself many things.”

Affiliations

- MIT Department of Civil and Environmental Engineering
- MIT Center for Computational Science and Engineering (CCSE)
- MIT Schwarzman School of Computer Science
- MIT Geospatial Data Center (GDC)
- MIT Auto-ID Laboratory
- MIT Center for Complex Engineering Systems (CCES)
- MIT Consortium for Improving Critical Infrastructure Cybersecurity (IC3)
- MIT System Design and Management Program

Areas of Interest and Expertise

- Information Technology,
- Cyber/Physical Security
- Web-Based Education Technology
- Large Scale Network Simulation
- GeoNumerics of Granular and Powder Systems
- Modern Software Architecting and Cyber Security
- Web Services and Distributed Computing
- Discrete Element Simulation and Analysis of Discontinua

Courses Taught with MIT Professional Education

- Applied Generative AI for Digital Transformation
- Blockchain: Disruptive Technology
- Cloud & DevOps: Continuous Transformation
- Data Leadership: Transforming the Corporation's Operations, Management, and Mindset to Leverage Data, AI, and Cloud Computing
- Digital Transformation: From AI and IOT to Cloud, Blockchain, and Cybersecurity

Course Outline

Module 1

Amazing AI: An Introduction to Artificial Intelligence for Business Leaders

1. Big Data and AI for Leaders
2. Parsing Our Physical World
3. ChatGPT and No-Code Models
4. AI Applications
5. Reinforcement Learning 101
6. Data Pipeline Automation

Module 2

Frameworks for Continuous Data Innovation

1. Frameworks for System Design and Operations
2. Data-First Companies
3. Decision-Making Frameworks
4. Agile Software Development Methodology
5. Organizing Around Modern Pipelines
6. Axiomatic Design
7. Design of Organizations

Module 3

IT Architecture and Querying Data

1. Internet and The World Wide Web
2. Client-Server Architecture
3. Structured Query Language (SQL) and Database Architecture
4. Architect Simplicity
5. Legacy Microsoft 365 With No-Code Power Platform
6. The API Problem and Apollo Graph

Module 4

The Importance of Data

1. Data
2. Transforming the Organization
3. The First Data Company
4. Data Explosion
5. Artisan vs. Factory

Module 5

Data Platforms and Database Design

1. What is a Data Platform?
2. Database Design
3. SQL for Leaders
4. Database Design Deep Dive

Module 6

Data Science Acceleration and the Data Platform

1. AI Factories
2. Data Platforms: Data Ingestion
3. Data Platforms: Data Reporting
4. The Modern Data Stack
5. Data Stacks Case Study: JetBlue
6. Modern Data Stack Patterns

Module 7

The Cloud

1. Cloud History
2. Cloud Journey
3. The Cloud and Data Leadership

Module 8

Ethics, Information Governance, and the Modern Data Organization

1. Ethics – AI Bias and Fairness Part I
2. Ethics – AI Bias and Fairness Part II
3. Data Governance and Compliance
4. Data Leadership
5. The Transformation of Data Technology



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Certificate of Completion

All participants who successfully complete **Data Leadership** will receive a MIT Professional Education Certificate of Completion. In addition, they will also earn 6.4 Continuing Education Units (MIT CEUs).

In order to obtain CEUs, participants must complete a required CEU confirmation form. CEUs are calculated based on the number of learning hours in each course.



The Continuing Education Unit (CEU) is defined as 10 contact hours of ongoing learning to indicate the amount of time they have devoted to a non-credit/non-degree professional development program. To understand whether or not these CEUs may be applied toward professional certification, licensing requirements, or other required training or continuing education hours, please consult your training department or licensing authority directly.



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Welcome to the gateway

to MIT knowledge
and expertise for
professionals
around the world.

155

Participants from
155 countries

≠ 1600

More than 1,600
online activities

92%

92% of Professional
Alumni rated
their experience
as outstanding

About MIT Professional Education

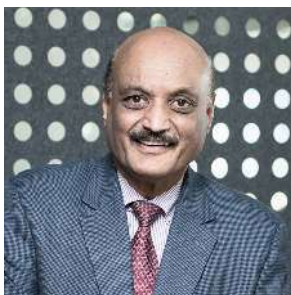
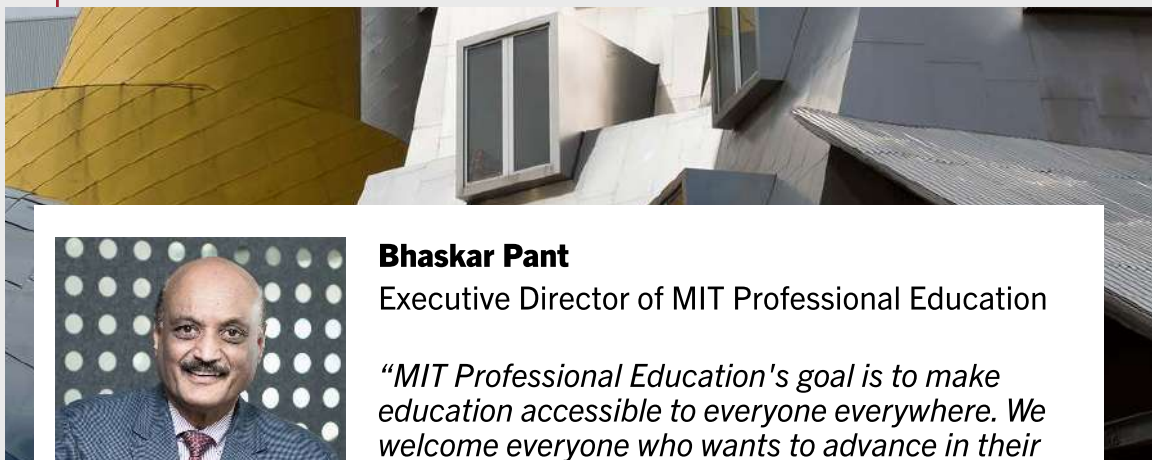
MIT Professional Education fosters the development of innovative leaders equipped to address complex problems globally

MIT Professional Education provides continuing education courses and lifelong learning opportunities for science, engineering, and technology professionals at all levels, from around the world. Worthy of note is that **MIT professors and lecturers lead and teach all MIT Professional Education offerings**. Some of our programs have a long history. Others are relatively new.

Our Digital Plus Programs go beyond online, blending innovative content delivered using the best of online technology and traditional classroom instruction in a flexible, collaborative learning environment.

MIT Professional Education is central to MIT's vision. It fulfills the mandate to connect practitioner-oriented education with industry and incorporate industry feedback and knowledge into MIT education and research.

The Institute is committed to generating, disseminating, and preserving knowledge while working with others to apply this **knowledge for the benefit of humankind**.



Bhaskar Pant

Executive Director of MIT Professional Education

"MIT Professional Education's goal is to make education accessible to everyone everywhere. We welcome everyone who wants to advance in their career or grow their business to unite and be part of the MIT Professional Education all-inclusive community."

Corporate Courses

At MIT Professional Education, we are committed to accompanying organizations as they aspire to provide their professionals with the ongoing education critical for **success in today's competitive environment**. That's why we offer plans for companies who wish to train their professionals with us.

Which program best suits your organization's needs?

1

Standard Program:

Teams or groups of at least 10 people will obtain special conditions in their registration for any of our programs.

2

Custom Programs:

Custom-made to meet the specific needs of each organization and its professionals.

3

Private Cohorts:

Available in Open Enrollment courses with a minimum of participants.



For more information on our Programs for Organizations, please contact us at: **professionalprograms@mit.edu**

The Beyond Online Methodology

MIT Professional Education is **revolutionizing the digital learning experience**. In fact, we are no longer talking about online-only, but rather an **interactive and collaborative learning experience** that's digitally powered: a **Beyond Online experience**.

You will have access to one of the most innovative e-learning platforms that utilizes the latest technology. You will also have an opportunity to meet and work with some of the world's leading subject matter experts in their respective fields.



With MIT Professional Education, you'll have an opportunity to:

- Attend select live webinars with MIT Instructors.
- Collaborate and engage with world-renowned MIT Instructors.
- Learn from expert learning facilitators, guest speakers, and subject matter experts from industries.
- Weekly live webinars from learning facilitators.
- Fulfill your reskilling and upskilling needs in this fast-changing environment.
- Master the technical and human skills necessary for effective leadership.
- Meet and network with other program participants from all around the world.



Clara Piloto

Director of Global Programs at MIT Professional Education

"Data powers everything that we do. The ability to speak the language of data opens unimaginable doors to new possibilities and opportunities for organizations. The combination of data with leadership exponentially strengthens your organization."

Benefits of Joining the Alumni Community



MIT Professional Education offers **a number of benefits for participants** who successfully complete our courses:

- 1 **15% discount** for MIT Professional Education online courses and short-duration, in-person programs.
- 2 **Continued access to the virtual campus for six months after completion** of the course.
- 3 **Exclusive announcements** of new courses, programs, and events through our **virtual Alumni Campus: Professional Network**.
- 4 Invitation to the **MIT Professional Education Alumni LinkedIn group**.
- 5 **Networking opportunities** with other MIT Professional Education Alumni.



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Are you ready to become a leader of data in your organization?

Frequently Asked Questions

For more information, please contact us:

+1 617 870 0270

Email: professionalprograms@mit.edu

Website: <https://professionalprograms.mit.edu>